Week 07: Embodiment, Situatedness and the Extended Mind

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Classical AI treated intelligence as reasoning over internal representations. This week examines the alternative that intelligence arises from bodies acting in environments: behaviour-based robotics, Braitenberg's synthetic psychology, enactivism and the thesis that the mind extends beyond the skull [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to explain Brooks's case against central representation and the subsumption architecture. They should be able to derive the behaviour of Braitenberg vehicles from their wiring and to describe the enactive and morphological views of cognition. Finally, they should be able to evaluate the parity argument for the extended mind.

Lecture

Intelligence without representation

Brooks argued that AI had focused on the wrong problems [1]. Instead of building disembodied reasoners with central world models, researchers should build complete creatures that act in the real world, starting with simple behaviours and adding layers. In his subsumption architecture, each layer connects perception directly to action, such as avoiding obstacles or wandering, and higher layers can suppress or override lower ones. The robots used the world itself as their source of information instead of maintaining detailed internal models.

Moravec observed a related asymmetry, often called Moravec's paradox [4]. Tasks that people find intellectually hard, such as chess, turned out to be comparatively easy for computers. Tasks that people find effortless, such as perceiving and moving, turned out to be very hard. Evolution spent far longer refining sensorimotor skills than abstract reasoning.

Synthetic psychology

Braitenberg described a series of imaginary vehicles, each with sensors connected to motors by simple wiring [2]. A vehicle with two light sensors, each driving the wheel on its own side more strongly when light is brighter, turns away from light and seems to fear it. Crossing the wires makes it turn towards the light and rush at it, as if aggressive. With inhibitory connections, the uncrossed vehicle approaches the light and slows down, facing it as if in love. The crossed one approaches and then turns away, like an explorer. Figure 7.1 shows these trajectories.

Braitenberg drew a methodological moral, the law of uphill analysis and downhill invention: It is easier to build a mechanism that produces a behaviour than to infer the mechanism from the behaviour. Observers readily attribute emotions and intentions to the vehicles, even though the mechanism is transparent to the designer. The interactive lab lets students rewire the vehicles and change their morphology, and Week 14 returns to the attribution of intentions.

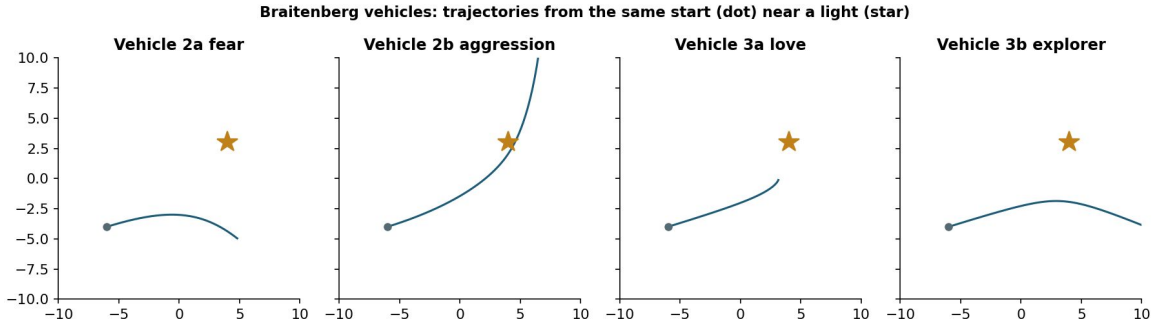


Figure 7.1. Trajectories of four Braitenberg vehicles that differ only in whether their sensor-motor connections are crossed and whether they excite or inhibit [2].

The embodied and enactive mind

Varela, Thompson and Rosch proposed that cognition is enaction [5]. Enaction is the bringing forth of a world through the history of structural coupling between an organism and its environment, grounded in sensorimotor capacities. Cognition, on this view, is not the representation of a pre-given world by a pre-given mind. Pfeifer and Bongard developed the engineering side of the idea [6]. They argued that the shape and material of a body take over part of the work that would otherwise require computation. They called this phenomenon morphological computation. A change in the angle of a vehicle's sensors, as in the interactive lab, can change its behaviour as much as a change in its wiring.

The extended mind

Clark and Chalmers asked where the mind stops and the rest of the world begins [3]. Their example contrasts Inga, who remembers the address of a museum, with Otto, who has a memory impairment and consults a Colab notebook he always carries. They proposed a parity principle. If a part of the world functions as a process that we would call cognitive if it occurred in the head, then that part of the world is part of the cognitive process. Otto's Colab notebook, on this view, is part of his memory.

The thesis raises questions about AI systems that rely on external tools and stores, and about people who rely on AI systems. Critics object that the Colab notebook lacks the integration and automatic trust that mark biological memory. The Colab notebook of this week simulates agents with limited internal memory that can consult an external store, which makes the parity principle a question about performance and cost.

> ***Pause and reflect.*** *Is a search engine you consult daily part of your memory in the sense of Clark and Chalmers? Which of their conditions does it meet?*

Hands-on components

Interactive lab: Choose the wiring of a two-sensor vehicle, move the light and change the angle of the sensors. Scrub through time to follow the vehicle. The label on each wiring is the emotion that Braitenberg suggested an observer would attribute [2, 6]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-07/lab.html

Colab notebook: It simulates Braitenberg vehicles and builds a small subsumption controller whose layers can be switched on and off. It also models an agent with limited internal memory that can consult an external store, in the spirit of the parity principle [1, 2, 3]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-07/NB07_embodiment.ipynb

Reflection prompts

1. Which Braitenberg vehicle surprised you most, and which descriptions came to mind before you checked the wiring?
2. Could a language model without a body be intelligent in the sense of Brooks or the enactivists? Give their likely answers and your own.
3. Identify a tool in your research practice that might count as part of your extended mind. Which conditions of Clark and Chalmers does it meet?

Weekly task and submission

What to do. Apply the embodied and extended views of cognition to one AI system of your choice. First, describe the body of the system, its sensors, its environment and the tools or stores it uses. Second, apply two views of this week, for example Brooks's case against central representation and the parity principle of the extended mind. Third, decide where the intelligence of the system is located: in the model, the body, the environment or their coupling. Use one result of the interactive lab or the Colab notebook as an example.

Where to look. Lecture: sections "Intelligence without representation", "Synthetic psychology", "The embodied and enactive mind" and "The extended mind". Interactive lab: the

Braitenberg vehicle, whose wiring and sensor angle you choose (tab: Explore). Colab notebook: Section 2 for the subsumption controller and Section 3 for memory inside and outside the head.

Example. A robot vacuum cleaner avoids walls mainly through its round body and its bump sensors, with little central planning. Under the parity principle, the map of the flat stored in its app could count as part of its cognitive process. Its intelligence would then lie in the coupling of controller, body and environment.

What to submit. About 500 words, and the Colab notebook with Exercises 1 and 2 completed. The midterm capstone, which also ends this week, is a separate piece of work.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Brooks, R. A. (1991). Intelligence without representation. Artificial Intelligence, 47(1-3), 139-159. https://doi.org/10.1016/0004-3702(91)90053-M

[2] Braitenberg, V. (1984). Vehicles: Experiments in Synthetic Psychology. MIT Press.

[3] Clark, A., & Chalmers, D. (1998). The extended mind. Analysis, 58(1), 7-19. https://doi.org/10.1093/analys/58.1.7

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[5] Varela, F. J., Thompson, E., & Rosch, E. (1991). The Embodied Mind: Cognitive Science and Human Experience. MIT Press.

[6] Pfeifer, R., & Bongard, J. (2006). How the Body Shapes the Way We Think: A New View of Intelligence. MIT Press.